

Wallcast PRD

Wallcast — PRD & Engineering Spec

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Wallcast is a Motif kit of ten image shaders for on-air commercial graphics, cyber-city screens and EDM video walls. You attach a still or a clip to a layer and the style treats it. Every style is loop-locked, reacts to audio through one mapping target, reads data through plain parameters, and stays inside the photosensitive limiter.

This document is both the product brief and the build spec. The kit it describes is built: wallcast-1.0.0.motifkit (17.7 KB), source in Motif3/kits/source/wallcast/.

1. Product overview

Problem. Broadcast, live-visual and signage teams start from stills: key art, product photography, hardware shots, poster art. They need those stills to move, stay on brand and stay legible at wall scale. Today that means After Effects plugin chains that do not run live, cannot be prompted, and do not loop on their own.

Vision. Drop an image, pick a look, hit the kick. Every look is a real-time shader that loops exactly, exports frame-accurately, and is safe to put on a wall.

Design principles

- 1. **Image first.** Each style is a treatment of your media, never a replacement for it. With nothing attached, a built-in skyline test image plays so a style is never blank.
- 2. **Loop exact.** Time enters only through the loop phase, in whole cycles. Frame 0 equals the last frame, and every frame renders on its own.
- 3. **One beat contract.** Every style has the same three beat controls, so audio mapping, presets and prompts work the same everywhere.
- 4. **Safe by default.** Every travelling band, flicker, tear, sparkle and pulse passes through the limiter. Styles that swing brightness carry the ⚡ mark.
- 5. **Data is a parameter.** Numbers reach the shader as ordinary keyable, mappable channels. Wallcast draws them as digits, meters and density.

Goals and success metrics (v1.0)

Goal	Metric	Target	Result
Ships as one installable kit	Installs in Motif 3.2 with no warnings	10 of 10 styles compile	Met
Works on any still or clip	Styles respond to attached media (mean pixel change vs. no media)	Above 0.5	Met, 51 to 66 on the SDK test card
Never blank	Styles render lit frames with no media	Coverage above 3%	Met, coverage 1.0
Loops exactly	Seam ratio between last and first frame	At or below 3	Met, worst 2.08 (Neon Trace)
Safe on a wall	Flashes per second in 25% of the frame at a 0.25 s worst-case loop	3 or fewer	Met, worst 3.5 in 1 of 16 tiles
Tunable	Parameters per style	4 or more	Met, 13 to 18
Real-time	Frame cost at 1080p on M-series	2.5 ms per style	To verify with Deliver › Check speed

2. Users and use cases

Persona	Job to be done	Key need
Broadcast designer	Turn campaign key art into a 6-second open with live numbers	Loop-exact export, safe flash rates, a matte for the headline
Live visual operator	Run a festival wall from one poster and a kick drum	Kick-locked motion, one mapping target, a wall preview
Environmental display designer	Ship content to a city-scale screen	Panel-pitch preview, long-run stability, no strobing
Prompt-first motion generalist	Say “tear it on the drop and glow the edges”	Readable parameter names, sane defaults, whole-kit consistency

Scenarios

- **Election-night open.** Key art on Depth Stack, a Contour Radar layer above it in Screen, and Slice Shift firing once on the stinger. The Readout parameter on the radar counts the tally.
- **Festival drop.** A poster on Kaleido Wall, Prism Split above it, Shockwave on top. The kick band maps to Hit on all three.
- **Night billboard.** A packshot on LED Wall at the panel’s pitch, Neon Trace for edge light, Glyph Mosaic with its Data feed mapped to a price channel.

3. Scope

In v1.0

- Ten styles, four palettes, one kit-level source media input (image or clip) and a second fore input on Depth Stack.
- Shared beat, finish and HUD controls on every style: 7 parameters, stamped by the generator.
- A procedural fallback image for every style.
- Seven-segment numerals in the shader for data readouts.
- Photosensitive-limiter routing for every time-driven term.
- Audit tooling that runs the kit inside the real Motif 3 app.

Out of v1.0

- Live audio uniforms in the shader. The SDK gives no audio to GLSL; audio reaches the shader through Motif’s band mapping onto parameters, which is why hit exists.
- A live data-channel API. The SDK has none, so data arrives as keyed or mapped parameters.
- Lottie export of shader passes. Shaders are raster; Motif bakes them to video or a PNG sequence.
- A third media input. The SDK allows two per style. A matte is a Motif layer mask, not a kit input.
- Custom fonts for Glyph Mosaic. The SDK has no texture inputs beyond media, so the glyph set is baked into the shader.

4. The ten styles

ID	Style	Group	Inputs	Params	Passes	Flash	Best for
prism-split	Prism Split	Optics	source	13	2	⚡	On-air, EDM wall
halftone-pulse	Halftone Pulse	Print and pixel	source	13	2		On-air, EDM wall
slice-shift	Slice Shift	Glitch and light	source	13	2	⚡	Cyber city, EDM wall
depth-stack	Depth Stack	Depth and data	source, fore	18	2		On-air, cyber city
contour-radar	Contour Radar	Depth and data	source	16	2		Cyber city, on-air
led-wall	LED Wall	Print and pixel	source	14	2	⚡	EDM wall, cyber city

ID	Style	Group	Inputs	Params	Passes	Flash	Best for
neon-trace	Neon Trace	Glitch and light	source	15	2	⚡	Cyber city, EDM wall
glyph-mosaic	Glyph Mosaic	Print and pixel	source	13	2		On-air, cyber city
shockwave	Shockwave	Optics	source	15	2	⚡	EDM wall, on-air
kaleido-wall	Kaleido Wall	Optics	source	16	2	⚡	EDM wall

Every style has two passes. Pass 0 (`styles/source.glsl`, half resolution) draws the fallback image and returns black at once when media is attached, so it costs nothing in normal use. Pass 1 is the treatment.

4.0 Shared controls

These seven parameters are stamped onto every style by `build/gen_manifest.py`. The shared library in `common.glsl` reads them, so it compiles in every pass.

Key	Control	Type	Range	Default	Group	Notes
beats	Beats / loop	int	1 to 16	4	Beat	Kick pulses per loop. The limiter caps this at 3 per second.
hitAmt	Beat drive	range	0 to 1.5	1.0	Beat	How hard the internal beat drives the effect.
hit	Hit	range	0 to 1	0.0	Beat	Map a live kick band here to drive the effect from audio.
exposure	Exposure	range	0.4 to 1.6	1.0	Finish	
vignette	Vignette	range	0 to 1	0.4	Finish	
hud	HUD frame	toggle	on / off	off	HUD	Corner brackets, numeric readout and a beat trace.
readout	Readout	range	0 to 9999	128	HUD	Shown as four digits. Key it or map a data channel to it.

`W_beat()` is a loop-locked kick envelope: $1.25 \cdot \text{smoothstep}(0, 0.04, f) \cdot \exp(-5f)$ with $f = \text{fract}(u_p \cdot r)$ and $r = \text{safeCycles}(\text{beats})$. `W_drive()` is `clamp(W_beat() * hitAmt + hit, 0, 1.5)`. Every style uses `W_drive()` for its beat response.

4.1 Prism Split

Radial chromatic split plus anamorphic light streaks. Red and blue sample along the radius from the frame centre; an optional seven-tap spectral fringe adds a rainbow edge; a 25-tap horizontal filter over highlights above 0.25 linear becomes an amber streak with a cool counter-streak. The kick widens the split and lengthens the streak.

Key	Control	Type	Range	Default	Group	Notes
split	Split	range	0 to 0.08	0.022	Style	
spectral	Spectral fringe	toggle	on / off	on	Style	Adds a seven-tap rainbow fringe on top of the RGB split.
streak	Streak gain	range	0 to 2	0.9	Streaks	
streakLen	Streak length	range	0.05 to 0.5	0.22	Streaks	

Key	Control	Type	Range	Default	Group	Notes
streakCool	Cool streak	range	0 to 1	0.4	Streaks	Cyan streak mixed in opposite the amber one.
sweeps	Light sweeps / loop	int	0 to 4	1	Streaks	

- **Loop.** The light sweep is `lsaw(safeCycles(sweeps))` with a $\sin(\pi \cdot t)$ envelope, so it fades at the frame edges and the seam closes.
- **Cost.** 35 media reads per pixel. The highest of the kit; the streak loop dominates.

4.2 Halftone Pulse

One halftone screen per colour channel at angles 0.26, 0.79 and 1.31 rad. Light mode adds RGB dots to the dark ground. Ink mode subtracts cyan, magenta and yellow from a paper ground. Dot radius follows a tone curve of the source and swells with the kick. The angle drift is $p_drift \cdot \text{lsin}(1)$.

Key	Control	Type	Range	Default	Group	Notes
cells	Screen cells	int	30 to 200	96	Style	Cells across the short side of the frame.
inkMode	Ink mode	select	light, print	light	Style	Light adds RGB dots to the ground. Ink subtracts CMY from paper.
gain	Dot gain	range	0.4 to 1.0	0.72	Style	
drift	Angle drift	range	0 to 0.15	0.03	Style	
registration	Registration	range	0 to 1	0.25	Style	Offsets the red and blue screens like a loose print run.
under	Source under-print	range	0 to 1	0.05	Style	Light mode only. Lets the original image show under the dots.

- **Loop.** Drift is one sine cycle per loop.
- **Cost.** 3 media reads per pixel. Uses screen-space derivatives for edge anti-aliasing.

4.3 Slice Shift

Row displacement, column-block jumps, an RGB tear and a scanline bed. Rows and blocks re-roll at `tslot(rate)`, so the limiter caps changes at 3 per second. The kick raises both the odds of a tear and how far it reaches.

Key	Control	Type	Range	Default	Group	Notes
bands	Bands	int	6 to 64	24	Style	
rate	Changes / loop	int	1 to 12	6	Style	Tear pattern changes per loop. The limiter caps it at 3 per second.
tear	Tear	range	0 to 1	0.55	Style	
blocks	Column blocks	range	0 to 1	0.4	Style	
rgb	RGB split	range	0 to 1	0.5	Style	
scan	Scanlines	range	0 to 1	0.5	Style	

- **Loop.** All randomness is hashed from the slot index and the layer seed.

- **Limiter.** With a loop shorter than one third of a second the slots freeze, and the tears hold still instead of strobing.
- **Cost.** 3 media reads per pixel.

4.4 Depth Stack

Background, a self-derived mid layer and a foreground input at three parallax depths. Luminance adds per-pixel depth to the background shift. The mid layer is a deeper, slightly enlarged copy of the background, screen-blended where it is bright. The foreground keeps its alpha, casts an offset shadow and can be luma-keyed when it is an opaque photo. A camera XY pad and a one-cycle orbit move all three.

Key	Control	Type	Range	Default	Group	Notes
camera	Camera	point	-0.5 to 0.5 (XY)	0, 0	Camera	Parallax offset. Key or map it for a camera move.
orbit	Orbit	range	0 to 1	0.5	Camera	Loops one slow circle per loop.
depthBg	Background depth	range	0 to 1	0.35	Depth	
depthMid	Mid depth	range	0 to 1	0.6	Depth	
depthFore	Foreground depth	range	0 to 1	1.0	Depth	
midBlend	Mid layer	range	0 to 1	0.7	Depth	Screen-blends a deeper copy of the background.
shadow	Shadow	range	0 to 1	0.55	Foreground	
lumaKey	Key out dark	toggle	on / off	on	Foreground	Drops near-black pixels of an opaque foreground photo.
keyLevel	Key level	range	0.005 to 0.2	0.03	Foreground	
sweeps	Light sweeps / loop	int	0 to 3	1	Style	

- **Inputs.** source is the background. fore is the foreground; transparent PNGs keep alpha. With no foreground, a procedural hardware slab plays.
- **Loop.** The orbit is $1c(1)$. The light sweep is limiter-routed.
- **Cost.** 5 media reads per pixel.

4.5 Contour Radar

Iso-luminance contour lines from a 3×3 blurred luminance field, drawn with derivative-based anti-aliasing. A radar sweep brightens the lines as it passes, range rings add scale, and a reticle prints the local level as two digits. Lines drift by whole major intervals per loop.

Key	Control	Type	Range	Default	Group	Notes
levels	Contour levels	int	4 to 40	18	Style	
major	Major interval	int	2 to 8	4	Style	
drift	Contour drift	int	-3 to 3	1	Style	Whole major intervals per loop, so the loop closes.
sweeps	Radar sweeps / loop	int	1 to 3	1	Radar	
trail	Sweep trail	range	1 to 12	5	Radar	
rings	Range rings	toggle	on / off	on	Radar	
showReticle	Reticle	toggle	on / off	on	Reticle	

Key	Control	Type	Range	Default	Group	Notes
reticle	Reticle position	point	-0.9 to 0.9 (XY)	0.15, 0.1	Reticle	Key or map it to follow a subject.

- **Loop.** Drift is $\text{sign} \cdot \text{floor}(\min(\text{drift} \cdot \text{major}, 3 \cdot L) / \text{major}) \cdot \text{major}$ levels per loop, so it stays a multiple of the major interval and no line crosses a pixel more than 3 times per second.
- **Cost.** 10 media reads per pixel, plus 9 in a disc around the reticle only.
- **Data.** Map the reticle position to follow a subject; map readout to a number.

4.6 LED Wall

A pixel-mapped panel. Cells high sets the pitch; each cell shows the source at its centre through square or round masks, RGB sub-pixel stripes, a gap control and eight-tap bloom. A scan wipe travels the panel and time-slotted sparkle lights single cells.

Key	Control	Type	Range	Default	Group	Notes
cells	Cells high	int	24 to 160	72	Style	Panel pitch. Set it to match the real wall.
shape	Cell shape	select	square, round	square	Style	
gap	Cell gap	range	0 to 1	0.4	Style	
subpixel	Sub-pixels	range	0 to 1	0.5	Style	
bloom	Bloom	range	0 to 1.5	0.5	Style	
sparkle	Sparkle	range	0 to 1	0.3	Style	Time-slotted, so the limiter caps its rate.
wipes	Scan wipes / loop	int	0 to 3	1	Style	

- **Loop.** The wipe is limiter-routed with a $\sin(\pi \cdot t)$ envelope. Sparkle uses `tslot(24)`.
- **Cost.** 9 media reads per pixel.
- **Use.** Set Cells high to match the real wall, then judge type and logos at that pitch before shipping.

4.7 Neon Trace

Sobel edge detection at three scales (line width, 4× and 10× the spread) drawn as light. The palette is a spectrum, the kit accents or mono ink, cycling a limiter-capped whole number of times per loop. A travelling gate brightens edges as it passes.

Key	Control	Type	Range	Default	Group	Notes
width	Line width	range	0.5 to 4	1.5	Style	
glow	Glow	range	0 to 2	0.9	Style	
spread	Glow spread	range	0.5 to 2	1.0	Style	
gate	Gate sweeps / loop	int	0 to 3	1	Style	
colorMode	Colour	select	spectrum, accents, mono	spectrum	Colour	
hueCycles	Hue cycles / loop	int	0 to 3	1	Colour	
hueSpan	Hue span	range	0 to 1	0.35	Colour	
base	Source under-glow	range	0 to 0.5	0.12	Colour	

- **Loop.** Hue cycles and gate sweeps both pass through `safeCycles`.
- **Cost.** 26 media reads per pixel. Second highest of the kit.

4.8 Glyph Mosaic

The image rebuilt from ten 5×5 bitmap marks ordered by density (blank, dot, colon, plus, 1, 7, 0, #, 8, block). Rain trails fall in each column at a whole number of cycles per loop, capped by the limiter. Live cells swap glyphs at `tslot(3)` and `tslot(10)`. The Data feed parameter shifts the density mapping, so a mapped number fills or thins the mosaic.

Key	Control	Type	Range	Default	Group	Notes
cols	Columns	int	40 to 160	96	Style	
rain	Rain speed	int	0 to 4	1	Style	Whole cycles per loop. 0 stops the rain.
flicker	Flicker	range	0 to 1	0.35	Style	Live cells swap glyphs. Rate is capped by the limiter.
feed	Data feed	range	0 to 1	0.5	Data	Map a data channel here. Density fills as it rises.
contrast	Contrast	range	0.5 to 2	1.1	Data	
tone	Tone	select	source, amber, ink	source	Data	

- **Loop.** Rain speed per column is $\text{floor}(\min(\text{rain} \cdot (1 + \text{floor}(3 \cdot h)), 3L))$ cycles per loop.
- **Cost.** 1 media read per pixel. The cheapest of the kit.

4.9 Shockwave

Each kick launches a refractive ring from the origin. Ring i has age $\text{fract}(u_p \cdot r) + i$, so at a beat boundary ring i takes over from ring $i-1$ with no jump. The ring displaces the image along the radius, splits colour, and adds an amber glow.

Key	Control	Type	Range	Default	Group	Notes
origin	Origin	point	-0.9 to 0.9 (XY)	0, 0	Style	Where the rings start. Key or map it.
speed	Ring speed	range	0.1 to 1.2	0.42	Style	
refract	Refraction	range	0 to 1.5	0.8	Style	
sharp	Ring sharpness	range	4 to 24	11	Style	
rings	Rings	int	1 to 3	3	Style	
chroma	Colour split	range	0 to 1	0.5	Style	
glow	Ring glow	range	0 to 1	0.25	Style	

- **Loop.** $r = \text{safeCycles}(\text{beats})$. When the limiter freezes the rate, the rings are skipped.
- **Cost.** 3 media reads per pixel.
- **Hit.** `hit` adds ring strength on top of the internal beat.

4.10 Kaleido Wall

A mirror-fold kaleidoscope. The frame is rotated by whole cycles per loop, folded into `segments` mirrored wedges, scaled by pattern scale and a one-cycle breathe, then sampled through a mirrored repeat. Drift moves the source by whole mirror periods.

Key	Control	Type	Range	Default	Group	Notes
segments	Segments	int	3 to 12	6	Style	
spin	Spin / loop	int	-2 to 2	1	Style	
scale	Pattern scale	range	0.3 to 1.5	0.9	Style	
breathe	Breathe	range	0 to 0.3	0.08	Style	

Key	Control	Type	Range	Default	Group	Notes
drift	Drift / loop	int	0 to 2	0	Style	Whole mirror periods per loop.
chroma	Colour split	range	0 to 1	0.5	Style	
center	Centre	point	-0.5 to 0.5 (XY)	0, 0	Style	Moves the point the pattern folds around.
seams	Seams	range	0 to 1	0.35	Style	

- **Loop.** Spin and drift are limiter-routed integer cycles. Breathe is $\lfloor \sin(1) \rfloor$.
- **Cost.** 3 media reads per pixel.
- **Note.** The key `scale` replaces `zoom`, which the SDK reserves for the layer.

5. Functional requirements

ID	Requirement	Pri	Status
WC-01	Ten styles in one motif-kit@1 package, id wallcast, semver 1.0.0	P0	Built, validated
WC-02	Kit-level source media input (image or clip) on every style; fore on Depth Stack	P0	Built
WC-03	Every style renders a lit frame with no media attached	P0	Built, audited
WC-04	Shared Beat group: beats, hitAmt, hit, on every style	P0	Built
WC-05	hit is the documented audio-mapping target; internal beat is loop-locked	P0	Built
WC-06	Time enters only through u_p in whole cycles; frame 0 equals the loop end	P0	Built, preview seam check
WC-07	All flicker, sweep, tear, sparkle, rain, hue and spin routed through the limiter	P0	Built, audited
WC-08	Styles that swing brightness carry flash: true	P0	Built (6 of 10)
WC-09	Data readout: readout parameter drawn as four seven-segment digits in the HUD	P0	Built
WC-10	Data-driven density and position: Glyph feed, Contour reticle, Shockwave origin	P0	Built
WC-11	Four palettes: Amber Signal, Ice Broadcast, Festival Magenta, Steel Mono	P0	Built
WC-12	4 or more parameters per style, groups and hints, show conditions where a control is conditional	P0	Built (13 to 18)
WC-13	Point parameters for camera, reticle, origin and centre	P1	Built
WC-14	Text or graphic matte through Motif layer masks	P0	Host feature; manual test in §11
WC-15	Clip inputs work through the SDK's media pipeline with frame-exact export	P0	Host feature; manual test in §11

ID	Requirement	Pri	Status
WC-16	Audit tool that installs the kit in the real app and checks compile, NaN, coverage and flashes	P1	Built
WC-17	Low-cost quality tier for High-cost styles	P1	Backlog (\$14)
WC-18	Optional depth-map input for Depth Stack	P2	Backlog; needs a third input

6. Kit architecture

```

wallcast/
  manifest.json      generated by build/gen_manifest.py; 10 styles, 4 palettes, 146 params
  common.glsl        shared library compiled into every pass
  styles/
    source.glsl      pass 0 (half res): fallback image, black when media is attached
    prism-split.glsl pass 1 of each style: the treatment
    halftone-pulse.glsl slice-shift.glsl depth-stack.glsl contour-radar.glsl
    led-wall.glsl     neon-trace.glsl glyph-mosaic.glsl shockwave.glsl kaleido-wall.glsl
  README.md
build/gen_manifest.py single source of truth for parameters
tools/wallcast-audit.mjs audit inside the real Motif 3 app

```

Pass graph. Every style: `source.glsl` (scale 0.5) → `<style>.glsl` (full size). The treatment reads the media through `m_source(q)` when attached, or `texture(u_buf0, q)` when not. The wrapper `w_src(q)` hides that choice.

Shared library (`common.glsl`)

Helper	Purpose
<code>w_src(q)</code>	Media composited on the palette background, or the fallback buffer
<code>w_beat()</code>	Loop-locked kick envelope through <code>safeCycles</code> (beats)
<code>w_drive()</code>	Beat times Beat drive, plus Hit; clamped to 1.5
<code>w_digit(p, d), w_num(uv, o, h, value, n)</code>	Seven-segment digits drawn in the shader
<code>w_hud(uv)</code>	Corner brackets, four-digit readout, beat trace with playhead
<code>w_finish(col, uv)</code>	Exposure and vignette

Why the generator. The SDK compiles `common.glsl` into every pass, and GLSL rejects a reference to an undeclared uniform. Stamping the seven shared parameters on all ten styles from one script keeps the kit compilable and the names identical.

SDK constraints that shaped the kit

Constraint	Effect
Two media inputs per style	Depth Stack uses source and fore; no matte input, masks are host-side
Reserved param names (zoom, rotate, phase, seed, tempo, loop, invert, palette)	Kaleido uses scale; validation caught this
Limit 32 params per style, 48 uniforms	Largest style uses 18
No state between frames	All motion is closed-form in <code>u_p</code> ; no trails or feedback
No audio in GLSL	Audio is host mapping onto <code>hit</code> and any other parameter
Media is premultiplied linear	<code>w_src</code> adds the palette background under transparent pixels

7. Audio, data and matte in Motif

Audio. Load a track in Motif and map its bands to parameters (amount and smoothing per map). Suggested maps:

Style	Parameter	Band	Amount	Smoothing
Any	hit	onset (kick)	0.8 to 1.0	Fast attack, 120 ms release
Prism Split	streak	high	0.5	80 ms
LED Wall	sparkle	high-mid	0.6	40 ms
Kaleido Wall	breathe	bass	0.15	150 ms
Neon Trace	glow	level	0.7	100 ms
Halftone Pulse	gain	bass	0.2	100 ms

Set hitAmt to 0 when the beat comes from audio only, so the internal clock does not double-hit. Set beats to the bar’s beat count when there is no track and the loop is the clock.

Data. Any number that should show on screen goes to a parameter: readout prints digits in the HUD of every style; Glyph Mosaic feed fills the mosaic; Contour Radar reticle follows a position; Shockwave origin moves the source of the rings. Key them, or map a band or a saved signal to them.

Matte. Put a Motif type layer or a shape under the Wallcast layer and use it as a track matte (alpha or luma), with feather and invert. The effect cuts out of the headline, the headline sits on the effect, or both. This is host behaviour; see the manual tests in §11.

8. Loop rule and determinism

Term	Where	Form
Beat envelope	all	$\text{fract}(u_p \cdot \text{safeCycles}(\text{beats})), \text{integer cycles}$
Light sweep, wipe, gate	Prism, Depth, LED, Neon	$\text{lsaw}(\text{safeCycles}(k))$ with $\sin(\pi \cdot t)$ or off-screen ends
Radar sweep	Contour	$\text{fract}(-\text{angle}/\tau - \text{safeCycles}(k) \cdot u_p)$
Contour drift	Contour	whole major intervals, capped at 3 line passes per second
Rain	Glyph	integer cycles per column, capped
Orbit, breathe, angle drift	Depth, Kaleido, Halftone	$\text{lc}(1), \text{lsin}(1)$
Spin, mirror drift	Kaleido	integer cycles through safeCycles
Hue cycles	Neon	integer cycles through safeCycles
Tears, flicker, sparkle	Slice, Glyph, LED	$\text{tslot}(n)$ slot index into seeded hashes
Ring age	Shockwave	$\text{fract}(u_p \cdot r) + i$

Randomness is always a hash of position, slot and the layer seed. Nothing reads wall-clock time. Exports render each frame from u_p alone.

9. Photosensitive safety

The limiter is on by default in the Kits tab. When on, safeCycles, tslot, tfrac and strobe cap changes at 3 per second based on the effective loop length, and freeze at zero when even one change per loop would exceed that.

Method. The audit renders each style at a worst-case loop of 0.25 s (a 1 s loop at 4× tempo), a 1 s loop and a 6 s loop, with default parameters and four random parameter sets. It reads 64×36 frames at 24 phases, splits them into a 4×4 grid of tiles, and counts pairs of opposing relative-luminance changes of 0.1 or more (with the darker frame below 0.8) per second in each tile. A style fails when 4 or more tiles (25% of the frame) exceed 3 per second.

Found and fixed. The first audit measured Contour Radar at 16 flashes per second in 14 tiles at the worst-case loop. The radar sweep and the contour drift ran at full speed. Both now go through the limiter, and contour drift is capped in whole major steps. The other styles’ sweeps, wipes, hue cycles, spin, drift and rain were routed the same way.

Result. Worst case across the kit: 3.5 per second in 1 of 16 tiles (Contour Radar). That is 6% of the frame against a 25% area rule, so it passes. It is very likely the HUD beat trace or the readout tile; if a stricter reading of the rule is needed, turn the HUD off.

The audit is an approximation of WCAG 2.3.1 and is not a Harding test. Content for broadcast still goes through the broadcaster’s own check.

10. Performance

Cost is media reads per pixel in the treatment pass, plus one half-resolution fallback pass that exits early with media attached.

Style	Media reads / px	Class	Note
Glyph Mosaic	1	Low	Table lookups only
Halftone Pulse	3	Low	Derivatives for edges
Slice Shift	3	Low	
Shockwave	3	Low	Loop of up to 3 rings
Kaleido Wall	3	Low	
Depth Stack	5	Low	Foreground read twice for the shadow
LED Wall	9	Mid	Eight-tap bloom
Contour Radar	10 (+9 near the reticle)	Mid	3×3 blur for stable lines
Neon Trace	26	High	Three Sobel scales
Prism Split	35	High	25-tap streak plus 7-tap fringe

Audit timings (CPU, SwiftShader, 480×270). Prism Split 84 ms, Halftone Pulse 23 ms, Slice Shift 19 ms, Depth Stack 28 ms, Contour Radar 44 ms, LED Wall 34 ms, Neon Trace 63 ms, Glyph Mosaic 18 ms, Shockwave 23 ms, Kaleido Wall 26 ms.

These are software-renderer numbers. They are useful for ranking styles on one machine and are not GPU times. Verify real cost in Motif with Deliver > Check speed at export size, and stay under 2.5 ms per style at 1080p on an M1.

Levers when a style is too heavy. Lower Neon Trace glow spread (fewer distinct fetches are not possible, but a smaller spread improves cache hits), turn Prism Split spectral fringe off (saves 7 reads), or lower the layer’s shader resolution. A half-tap quality tier is in the backlog.

11. Testing and acceptance

Automated (run in this drop)

Check		Tool		Result		
Schema and static checks		motif-kit validate		ok: 10 styles, 4 palettes, 146 params, 10 with media input		
Compile, loop seam, blank-frame, media response		motif-kit preview on the SDK test card and on four stills		All ✓; worst seam ×2.08; media Δ 51 to 66		
Install in the real app, compile every pass		__m_kits.install in Motif3.html		ok, no warnings		
NaN, Inf and coverage over 11 parameter sets × 4 palettes × 3 phases		tools/wallcast-audit.mjs		Coverage 1.0, NaN 0 for all ten		
Flash count at worst-case loop		tools/wallcast-audit.mjs		Passes; see §9		
Style	Params	Coverage	NaN share	Worst flashes / s in any tile	Tiles over 3 / s (of 16)	ms per 480×270 frame (CPU)
Prism Split	13	1	0	0	0	84
Halftone Pulse	13	1	0	0	0	23

Style	Params	Coverage	NaN share	Worst flashes / s in any tile	Tiles over 3 / s (of 16)	ms per 480×270 frame (CPU)
Slice Shift	13	1	0	0	0	19
Depth Stack	18	1	0	0	0	28
Contour Radar	16	1	0	3.5	1	44
LED Wall	14	1	0	0.42	0	34
Neon Trace	15	1	0	1	0	63
Glyph Mosaic	13	1	0	0.33	0	18
Shockwave	15	1	0	0	0	23
Kaleido Wall	16	1	0	0.5	0	26

Manual (not yet run; needs Motif open on a GPU)

1. Drop wallcast-1.0.0.motifkit on Motif. Confirm ten styles appear with the MEDIA badge and library thumbnails show the skyline.
2. Attach each of the four reference stills to each style. Confirm no black frames and no visible loop seam.
3. Attach a clip. Confirm Fit to loop gives a seamless loop and export seeks frames deterministically.
4. Map onset to hit on Prism Split and Shockwave with a 128 BPM track. Confirm the flare and ring land on the kick.
5. Put a type layer under Slice Shift as a luma track matte. Confirm the tear appears only inside the letters. Repeat with invert.
6. Key readout from 0 to 9999 over the loop on Contour Radar. Confirm the digits count and never clip.
7. Export MP4, WebM and a PNG sequence of Depth Stack at 1080p. Confirm frame 0 equals the last frame and Check speed stays under budget.
8. With the limiter off, confirm ∞ styles can exceed 3 per second, then turn it back on.

12. Chains

A chain is a stack of Motif layers, each with its own style, blend mode and mask. Up to four layers.

Election-night open (6 s loop, 25 fps)

Layer	Style	Blend	Key settings
1	Depth Stack	Normal	Background = key art, orbit 0.5, depthFore 1.0
2	Contour Radar	Screen	levels 14, major 4, readout keyed to the tally, opacity 0.6
3	Slice Shift	Normal, masked	rate 6, tear 0.7, mask animates in for the last second only

Festival drop (8 bars)

Layer	Style	Blend	Key settings
1	Kaleido Wall	Normal	segments 6, spin 1, poster as source
2	Prism Split	Screen	hit mapped to onset, streakLen 0.3
3	Shockwave	Add	hit mapped to onset, rings 3, refract 0.6

Night billboard (loop, panel pitch)

Layer	Style	Blend	Key settings
1	LED Wall	Normal	cells set to the panel’s vertical pixel count, packshot as source

Layer	Style	Blend	Key settings
2	Neon Trace	Screen	colorMode accents, glow 0.7, opacity 0.5
3	Glyph Mosaic	Add, masked	tone amber, feed mapped to the price channel, mask to a lower band

13. Delivery and repo layout

Path	What
Motif3/kits/wallcast-1.0.0.motifkit	The installable kit
Motif/motifKits/wallcast-1.0.0.motifkit	Copy beside the other packaged kits
Motif3/kits/source/wallcast/	Editable source: manifest, common, styles, README, build/gen_manifest.py
Motif3/tools/wallcast-audit.mjs	Audit inside the real app
Motif/docs/Wallcast-PRD-and-Engineering-Spec.md	This document (also .docx, .pdf)

Versioning. The kit id wallcast and every style id and parameter key are stable, because presets and saved projects refer to them (wallcast/prism-split). Add parameters freely. Never rename or remove one; deprecate in the label instead. Installing a kit with the same id updates it.

Rebuild

```
python3 build/gen_manifest.py wallcast/manifest.json
node motif-kit-sdk/bin/motif-kit.mjs validate wallcast
node motif-kit-sdk/bin/motif-kit.mjs preview wallcast --media photo.jpg
node motif-kit-sdk/bin/motif-kit.mjs pack wallcast --out dist
node tools/wallcast-audit.mjs wallcast.bundle.json Motif3.html audit.json
```

14. Milestones and backlog

#	Milestone	Scope	Exit	Status
K0	Contract	Shared params, common library, fallback image, generator	Kit compiles in every pass	Done
K1	Ten styles	All ten treatments with limiter routing	Validate, preview and app install pass	Done
K2	Audit	Real-app audit with NaN, coverage and flash counts	Contour Radar fixed; all pass	Done
K3	In-app QA	The manual list in §11 on a GPU	All eight checks signed off	Next
K4	Presets	Saved looks per style and the three chains in §12	Chains reload from a look	Next
K5	v1.1	Quality tier, depth map, Field Warp, Data Bars	See below	Backlog

Backlog

- **Quality tier.** A quality select on Prism Split and Neon Trace that halves the tap count.
- **Depth-map input.** A third media input would need an SDK change; alternatively pack depth into the foreground's alpha.
- **Field Warp.** A flow-line gravity lens after the particle-warp reference: lines bend around an origin point and a radius set by the kick. Closed-form in u_p , no state.
- **Data Bars.** Bar and sparkline overlays driven by a set of mapped parameters.
- **Glyph sets.** Additional baked glyph tables (katakana-style, hex, binary) as a select.
- **Label text.** Small baked letter set so the HUD can print units beside the readout.

15. Risks and open questions

Risk	Impact	Mitigation
Software-renderer timings misread as GPU cost	Wrong optimisation work	Label them; measure with Deliver > Check speed on the target machine
Prism Split and Neon Trace heavy on iPad and phone	Dropped frames	Lower shader resolution; quality tier in v1.1
Flash approximation misses a case	Photosensitive risk on a wall	Limiter on by default; broadcaster check before air; HUD off for strict readings
Sobel and contour passes amplify film grain	Line crawl	Contour uses a 3×3 blur; add a pre-blur to Neon Trace if a source is noisy
Two-input SDK limit	No matte or depth input	Use Motif masks; revisit with SDK 1.3
Data has no live channel	Manual keying for numbers	Map a band or saved signal to readout and feed; ask for a data input in the SDK
WebGL 2 required	Older browsers fail to install	Same as the rest of Motif 3

Open questions

1. Does Motif's track matte support a type layer directly, or does it need a precomp?
2. Should the kit ship saved looks for the three chains, or leave that to the studio's look library?
3. Is a quality control acceptable as an SDK-wide convention across kits?
4. Should the HUD text set grow a baked alphabet so labels can sit beside the digits?